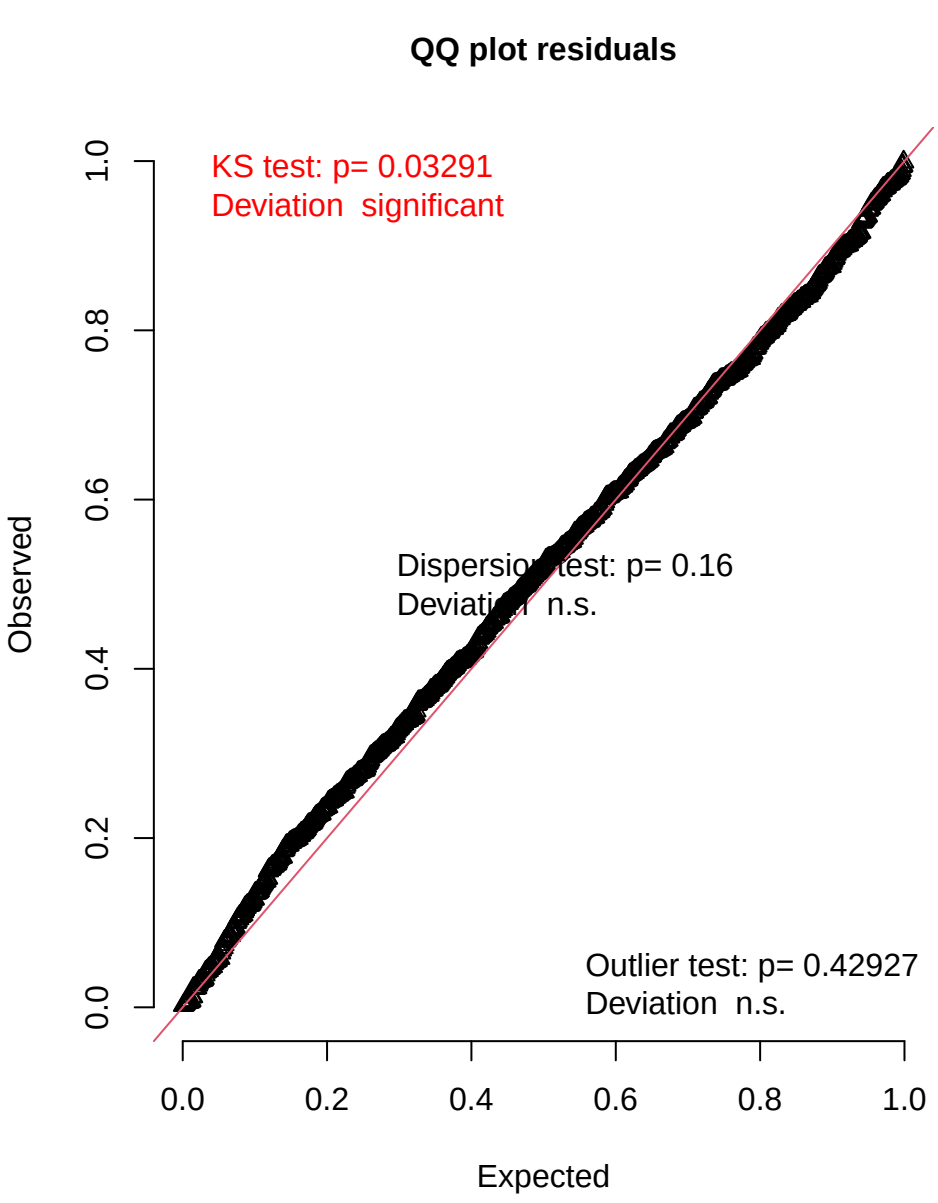
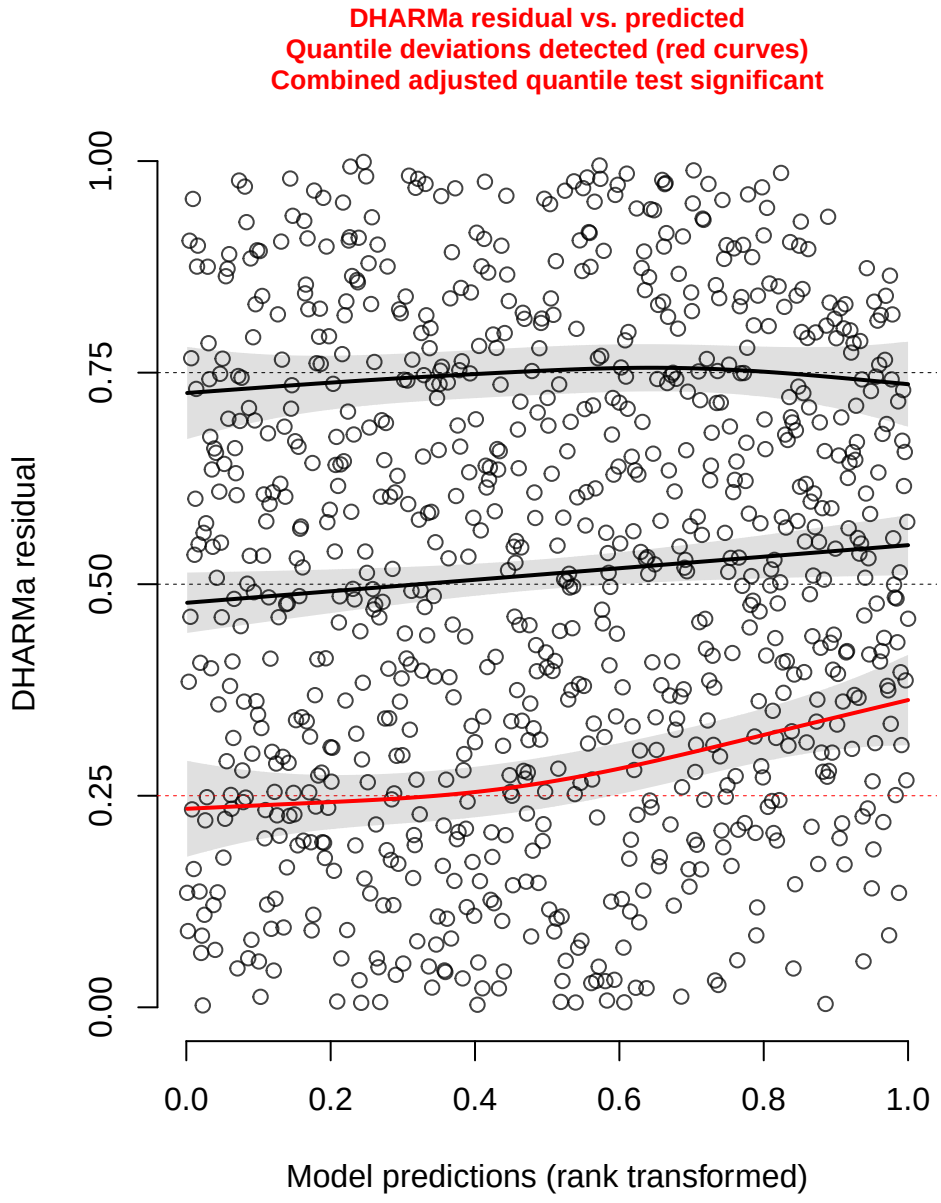
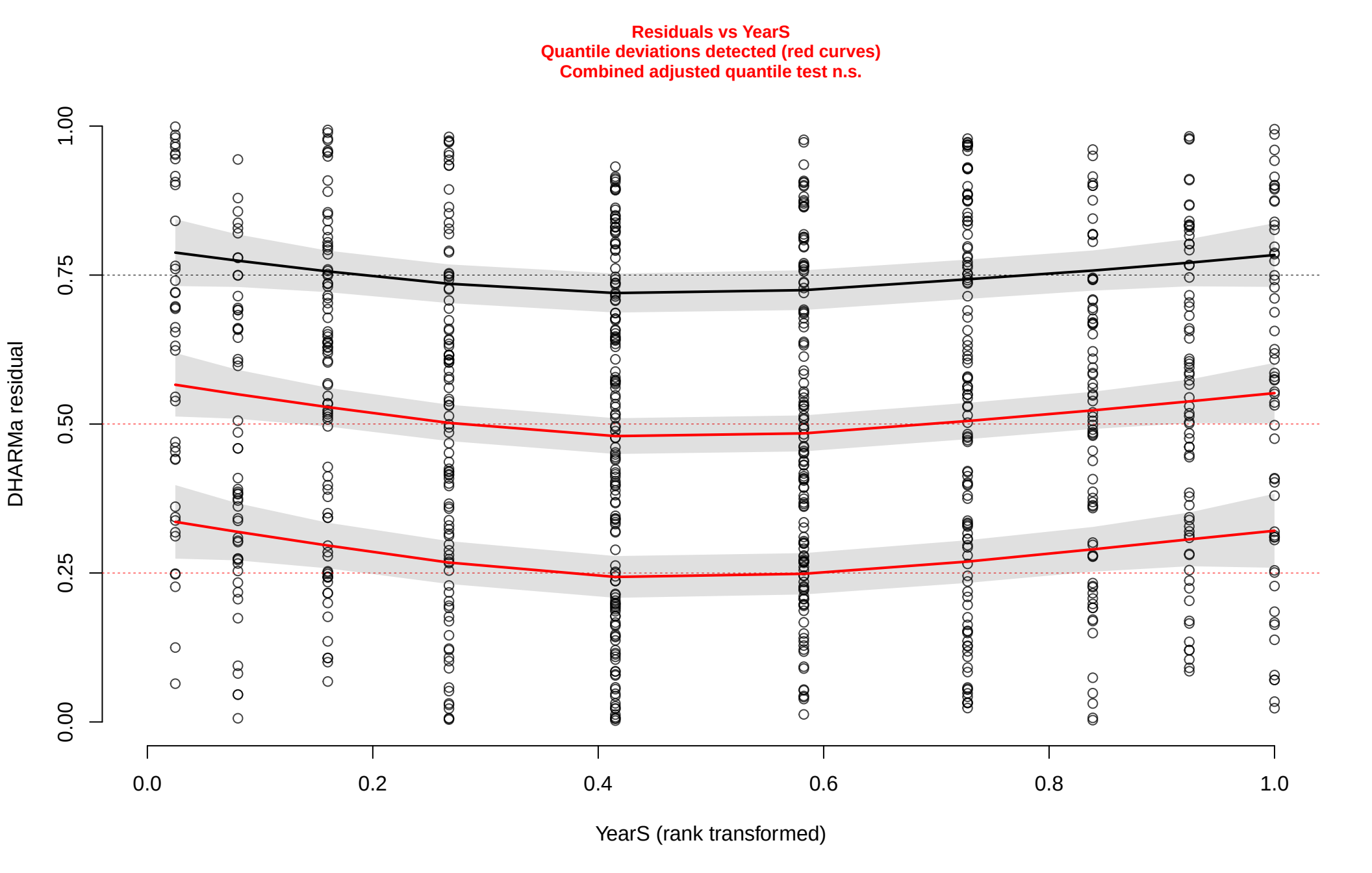
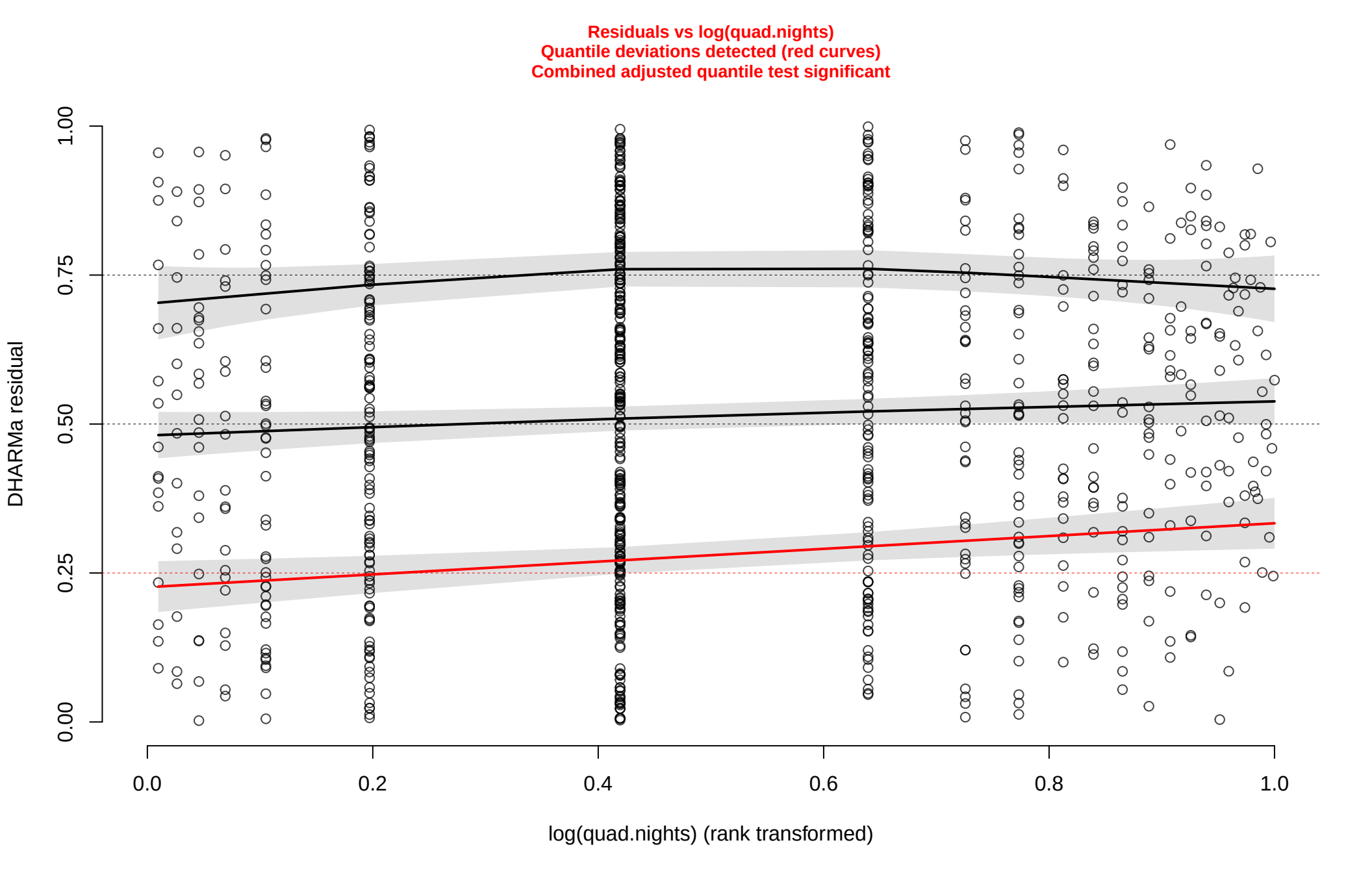
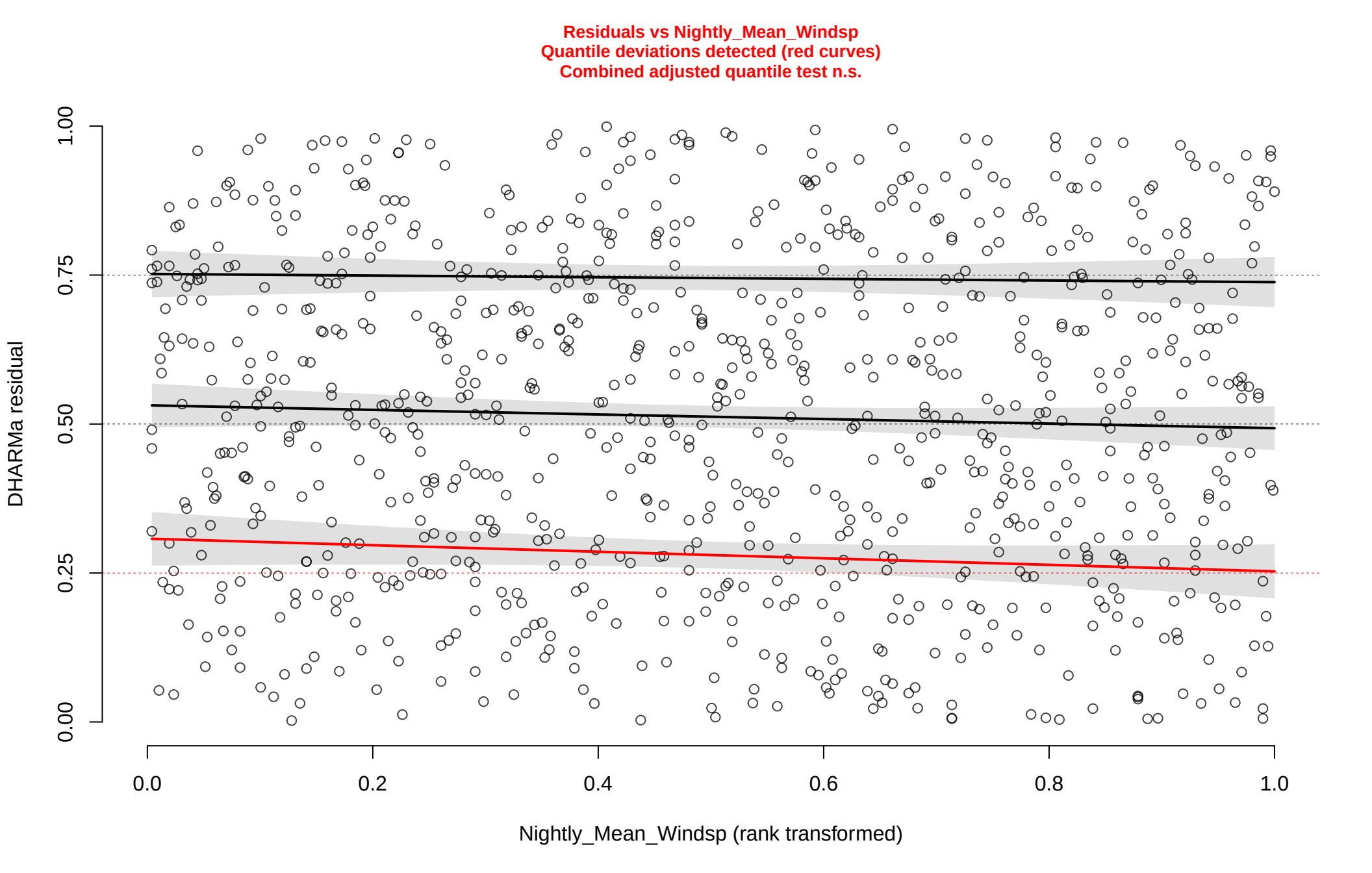
DHARMa residuals: EPFU

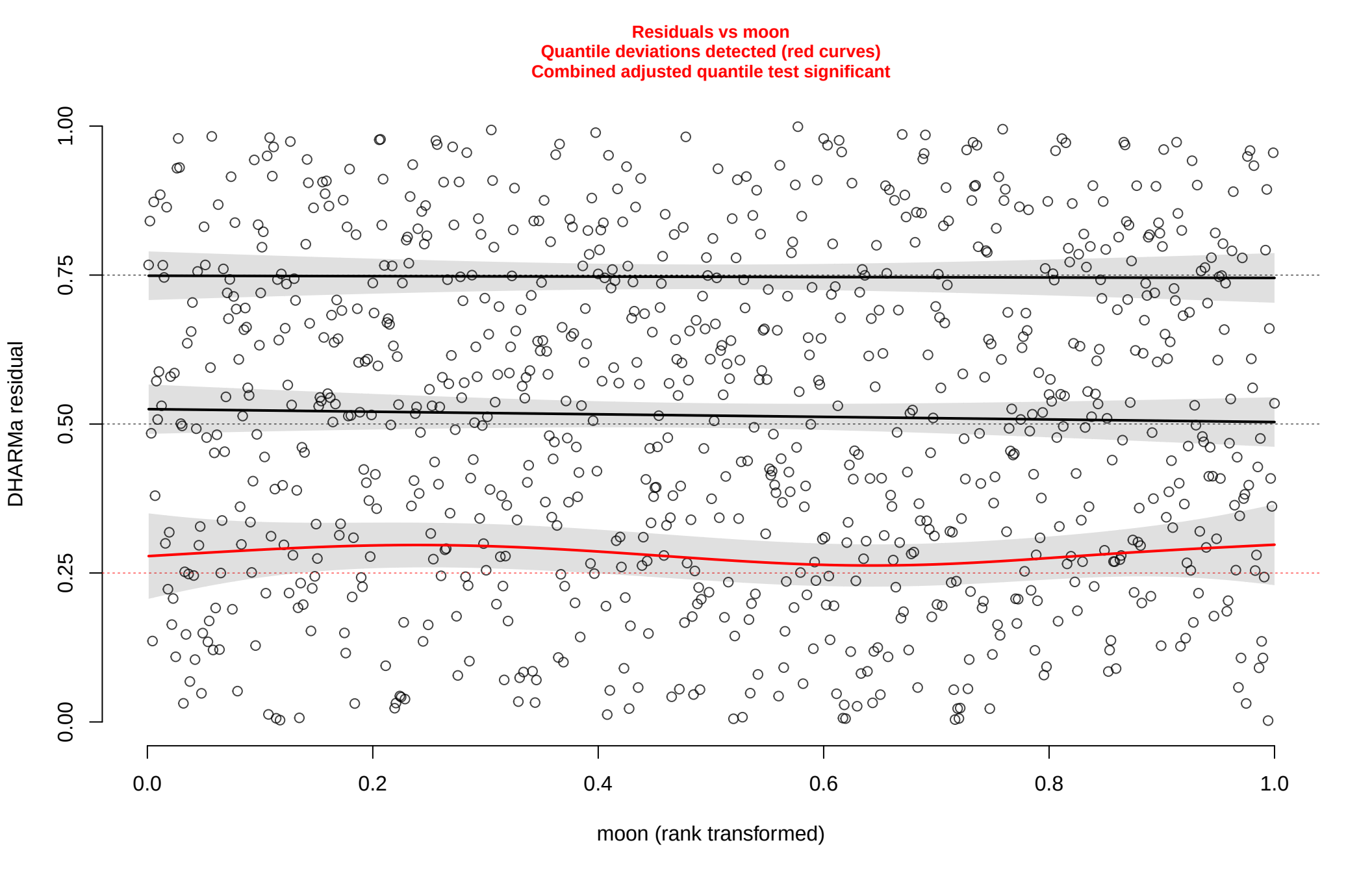


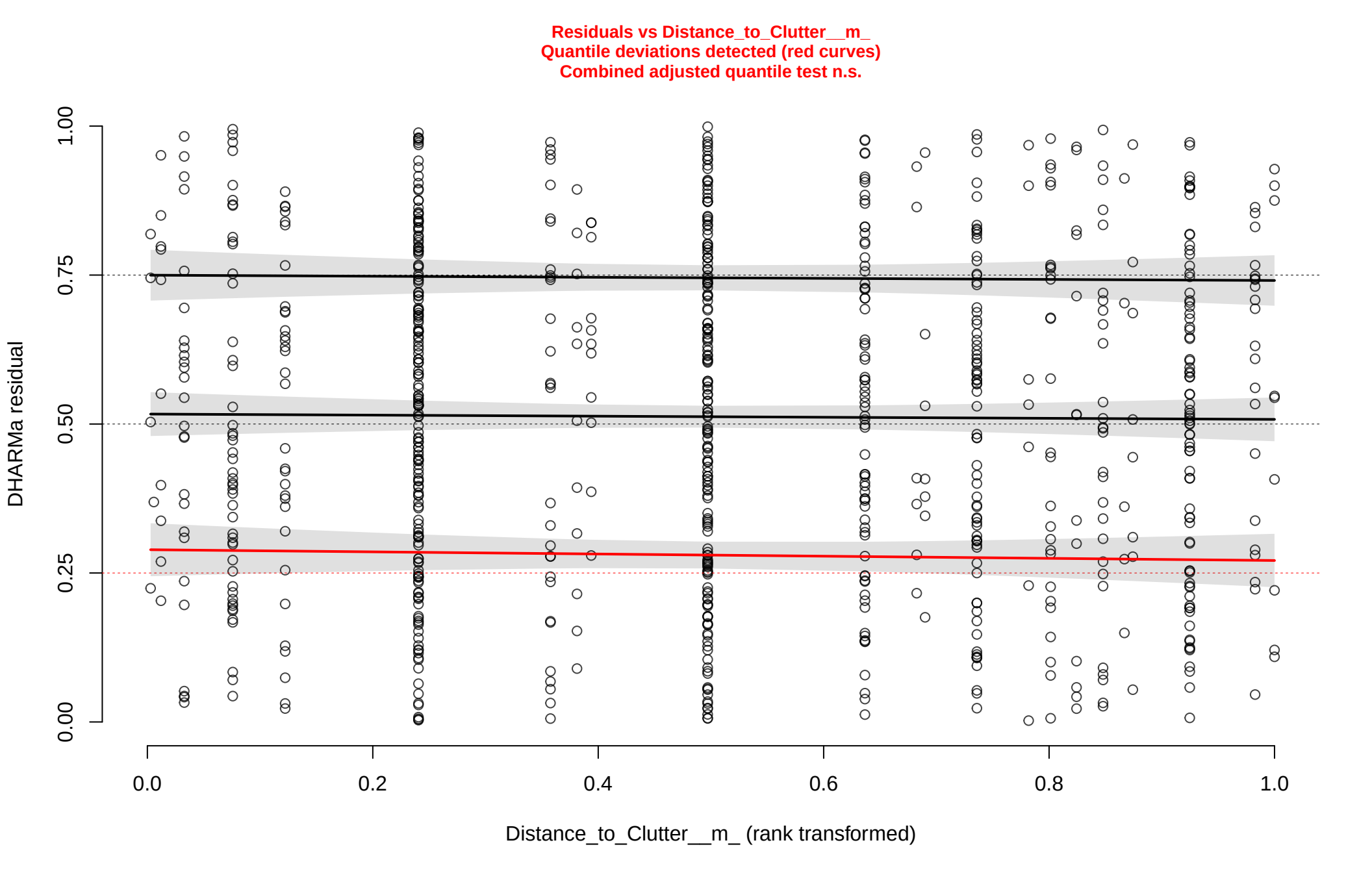


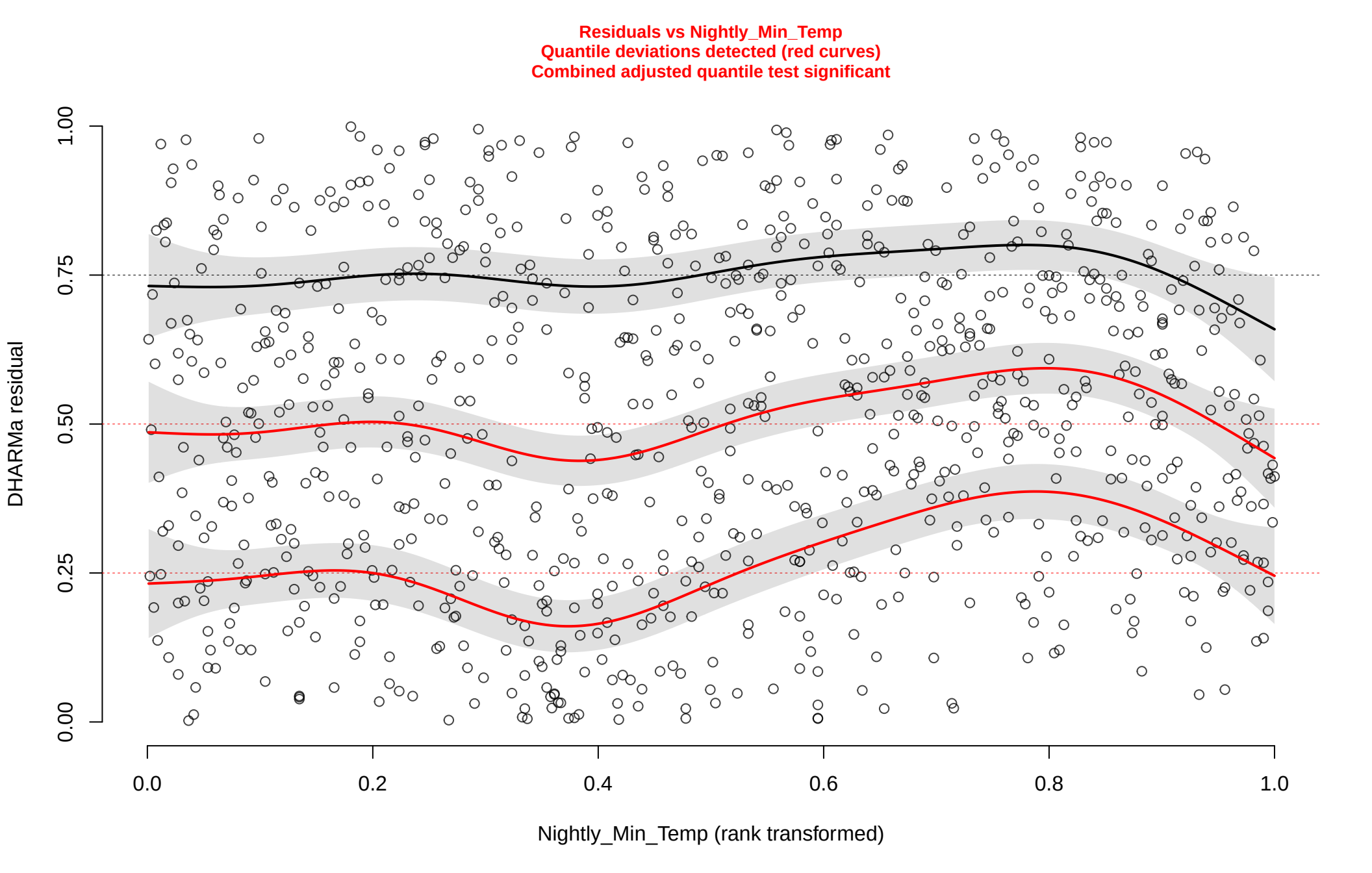

Residuals vs YearS
Quantile deviations detected (red curves)
Combined adjusted quantile test n.s.
DHARMa residual
1.00
0.75
0.50
0.25
0.00
0.0
0.2
0.4
0.6
0.8
1.0
YearS (rank transformed)

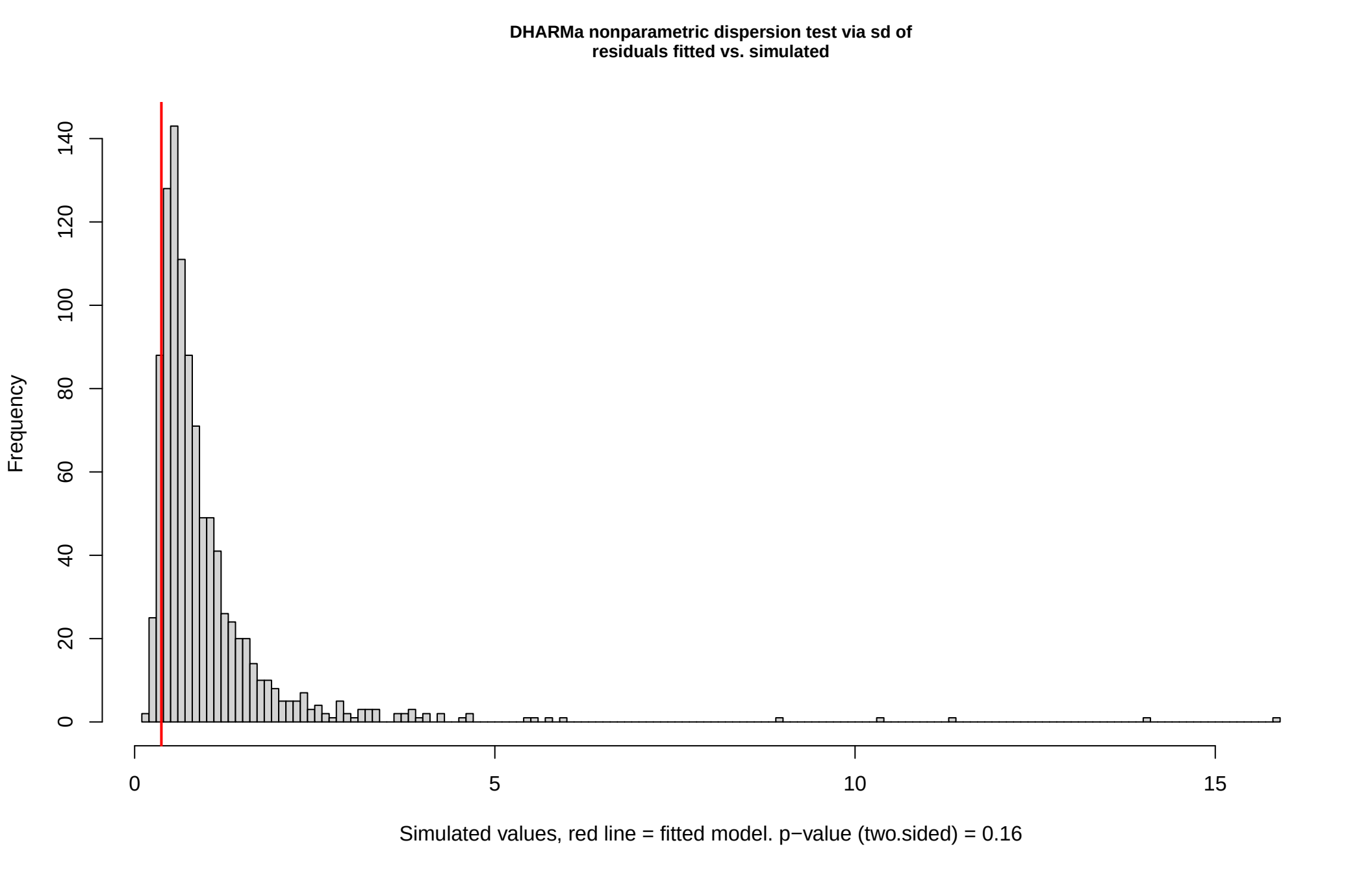

Residuals vs log(quad.nights)
Quantile deviations detected (red curves)
Combined adjusted quantile test significant
DHARMa residual
1.00
0.75
0.50
0.25
0.00
0.0
0.2
0.4
0.6
0.8
1.0
log(quad.nights) (rank transformed)

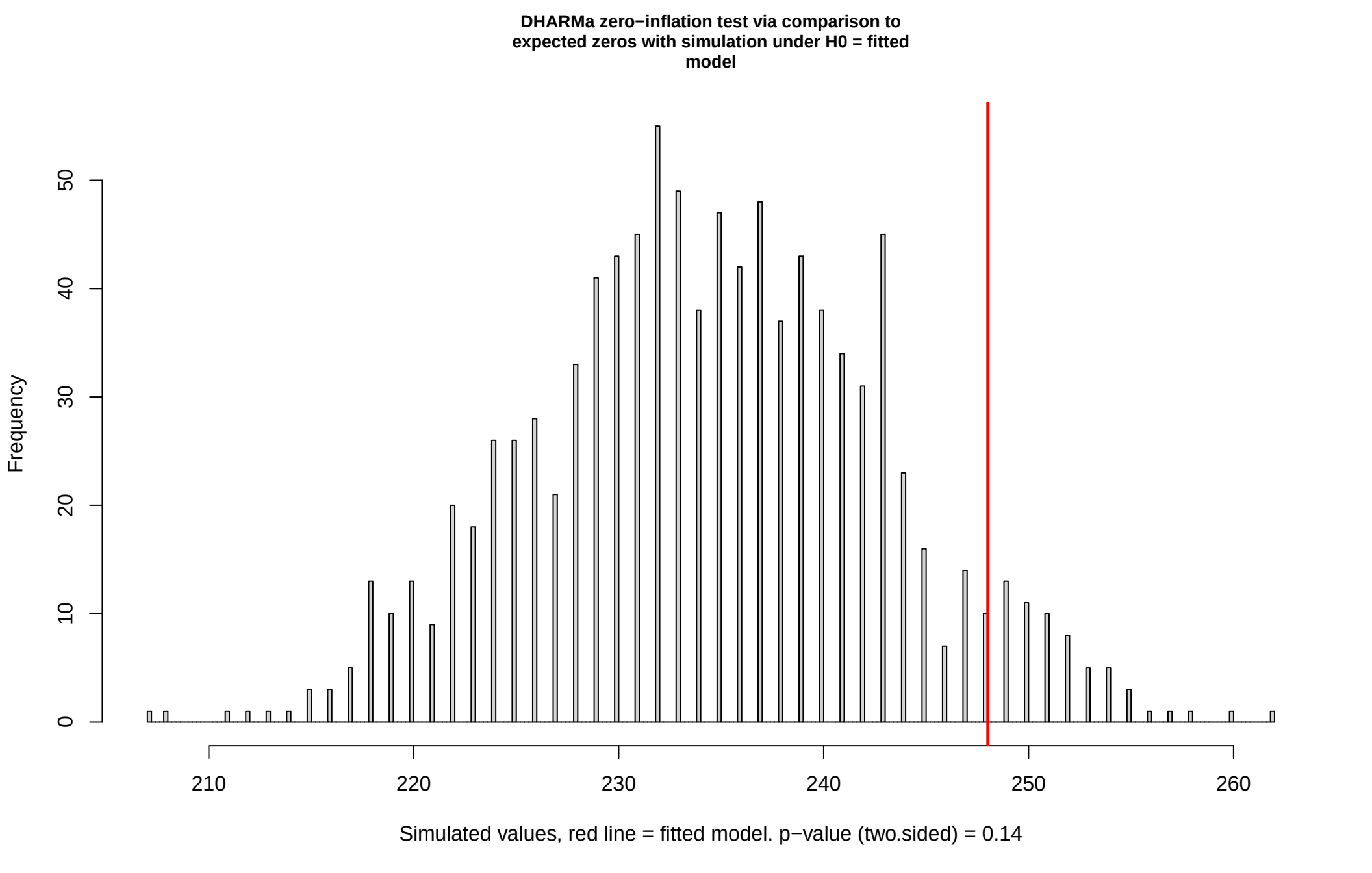

Residuals vs Nightly_Mean_Windsp
Quantile deviations detected (red curves)
Combined adjusted quantile test n.s.
DHARMa residual
0.00
0.25
0.50
0.75
1.00
0.0
0.2
0.4
0.6
0.8
1.0
Nightly_Mean_Windsp (rank transformed)

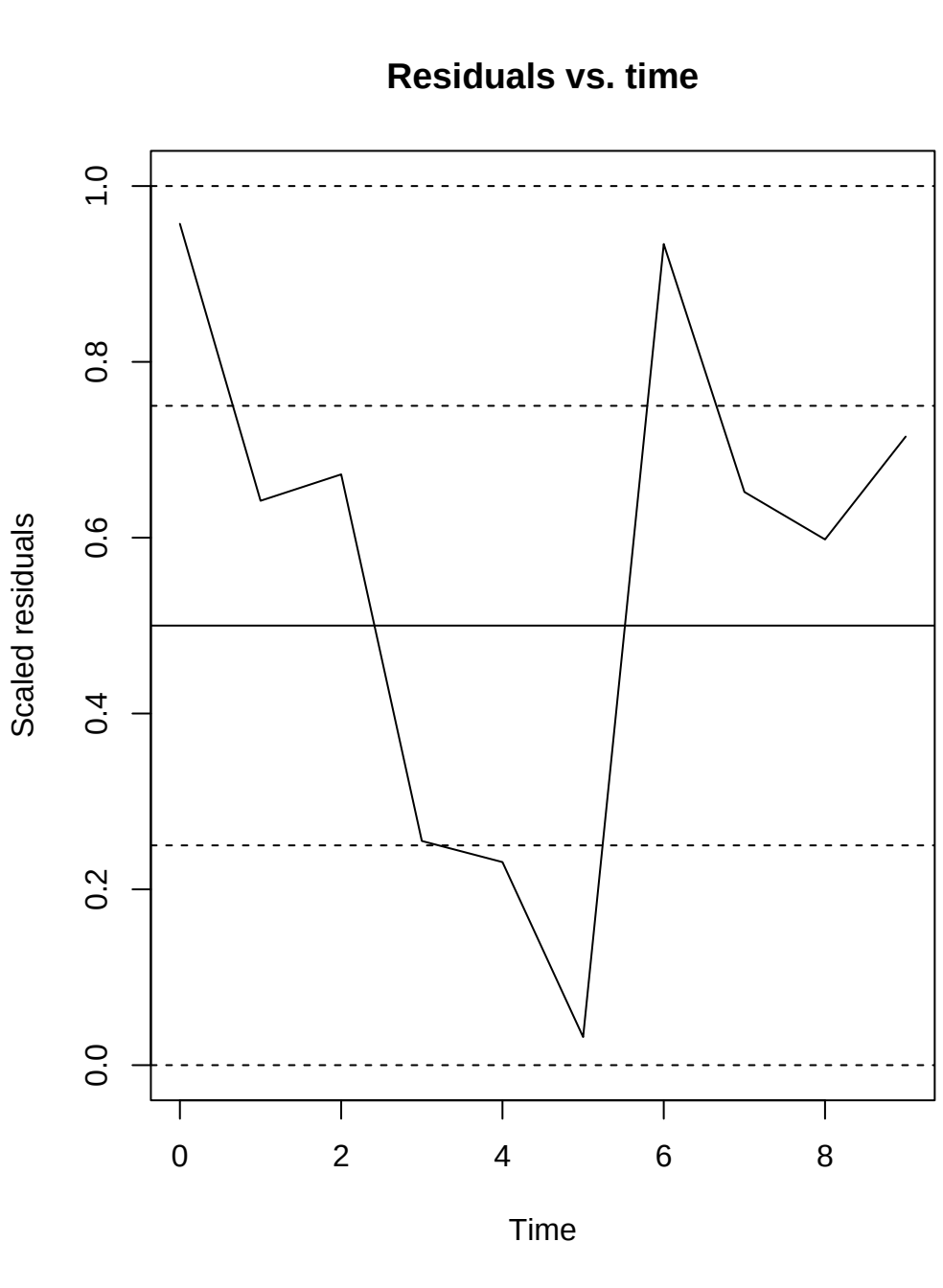

Residuals vs moon
Quantile deviations detected (red curves)
Combined adjusted quantile test significant
DHARMa residual
1.00
0.75
0.50
0.25
0.00
0.0
0.2
0.4
0.6
0.8
1.0
moon (rank transformed)

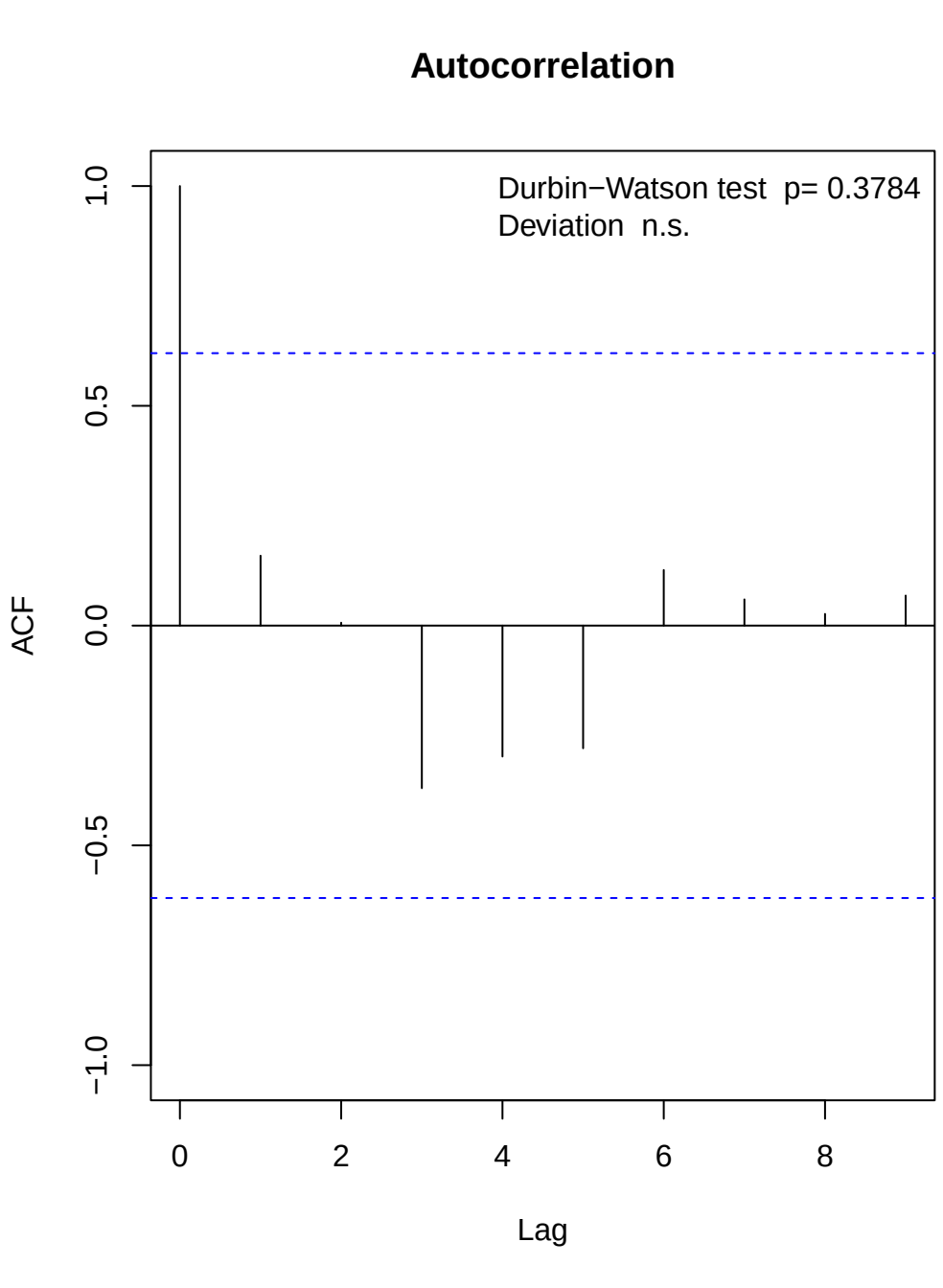

Residuals vs Distance_to_Clutter__m_
Quantile deviations detected (red curves)
Combined adjusted quantile test n.s.
DHARMa residual
1.00
0.75
0.50
0.25
0.00
0.0
0.2
0.4
0.6
0.8
1.0
Distance_to_Clutter__m_ (rank transformed)

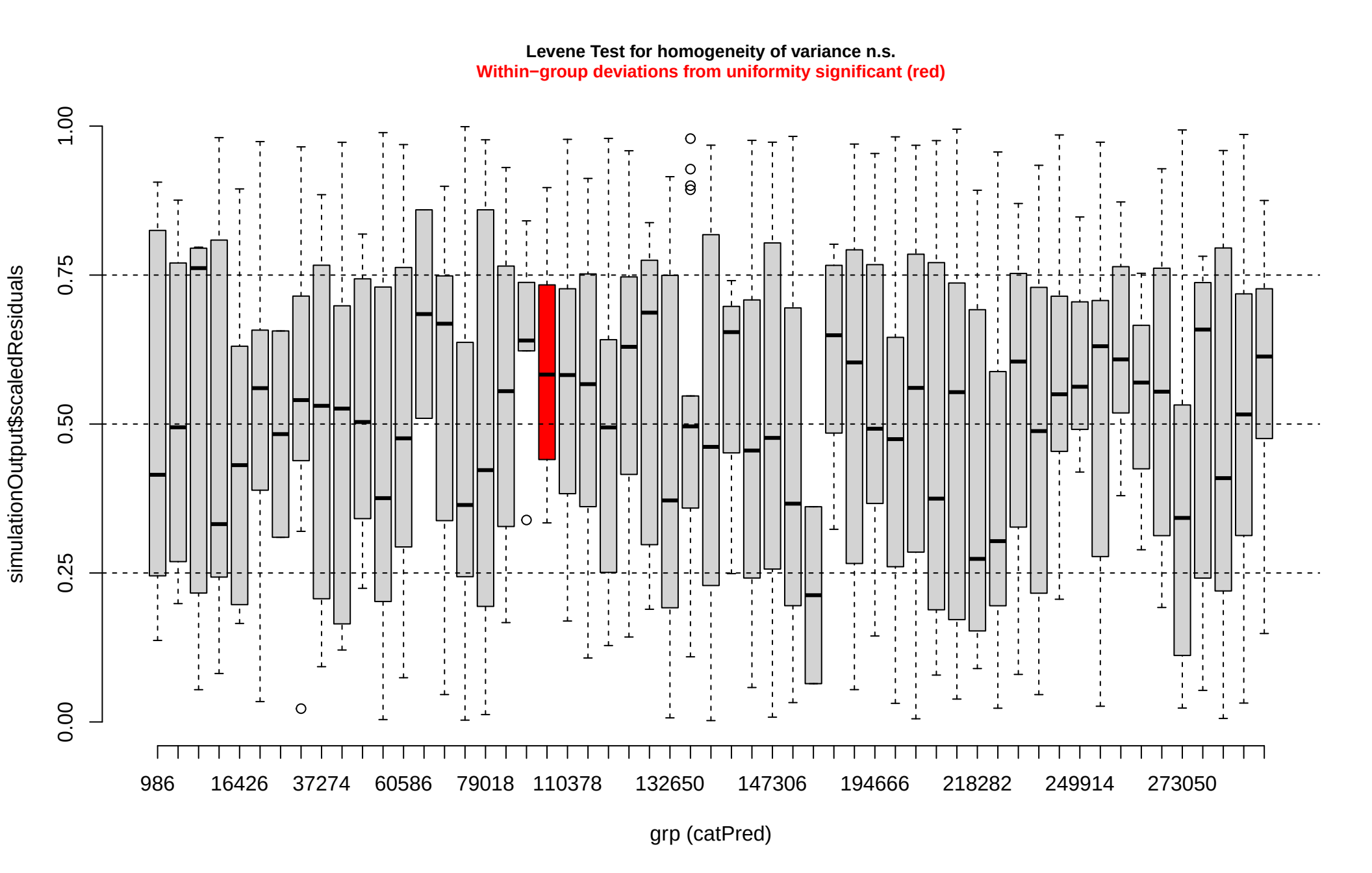

Residuals vs Nightly_Min_Temp
Quantile deviations detected (red curves)
Combined adjusted quantile test significant
DHARMa residual
1.00
0.75
0.50
0.25
0.00
0.0
0.2
0.4
0.6
0.8
1.0
Nightly_Min_Temp (rank transformed)

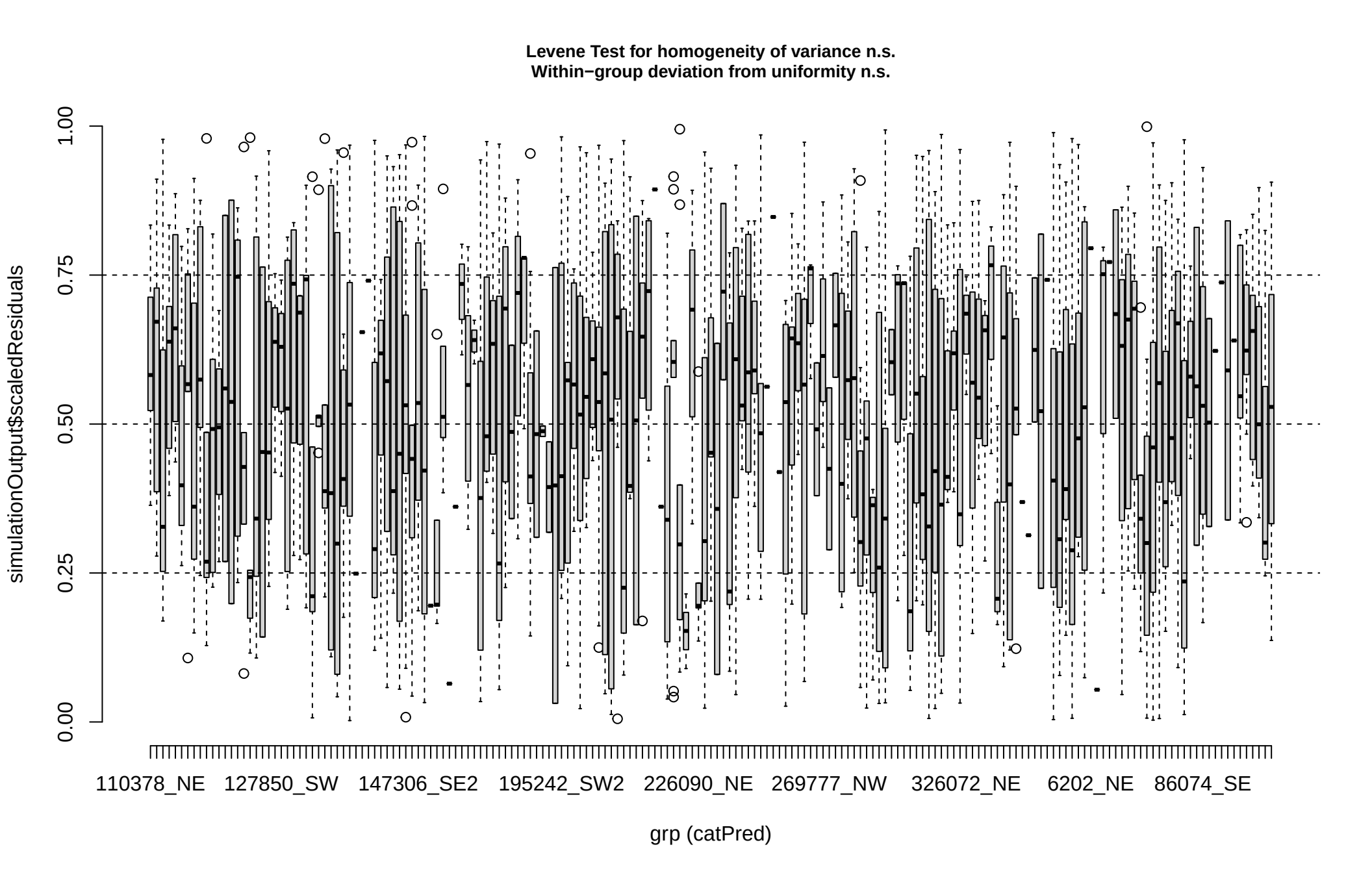

DHARMa nonparametric dispersion test via sd of residuals fitted vs. simulated
Frequency
0
20
40
60
80
100
120
140
0
5
10
15
Simulated values, red line = fitted model. p−value (two.sided) = 0.16

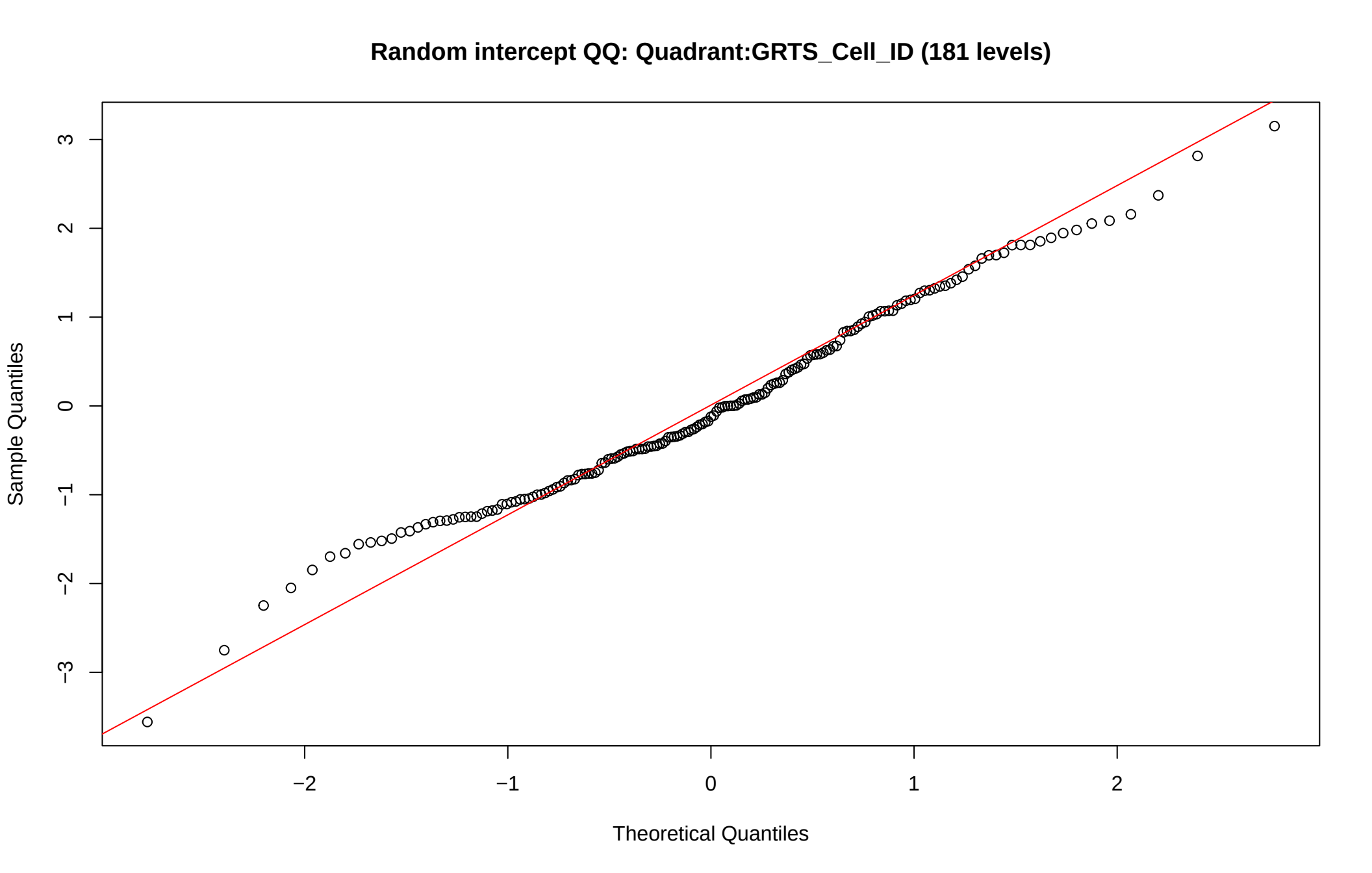

DHARMa zero−inflation test via comparison to expected zeros with simulation under H0 = fitted model
Frequency
0
10
20
30
40
50
210
220
230
240
250
260
Simulated values, red line = fitted model. p−value (two.sided) = 0.14

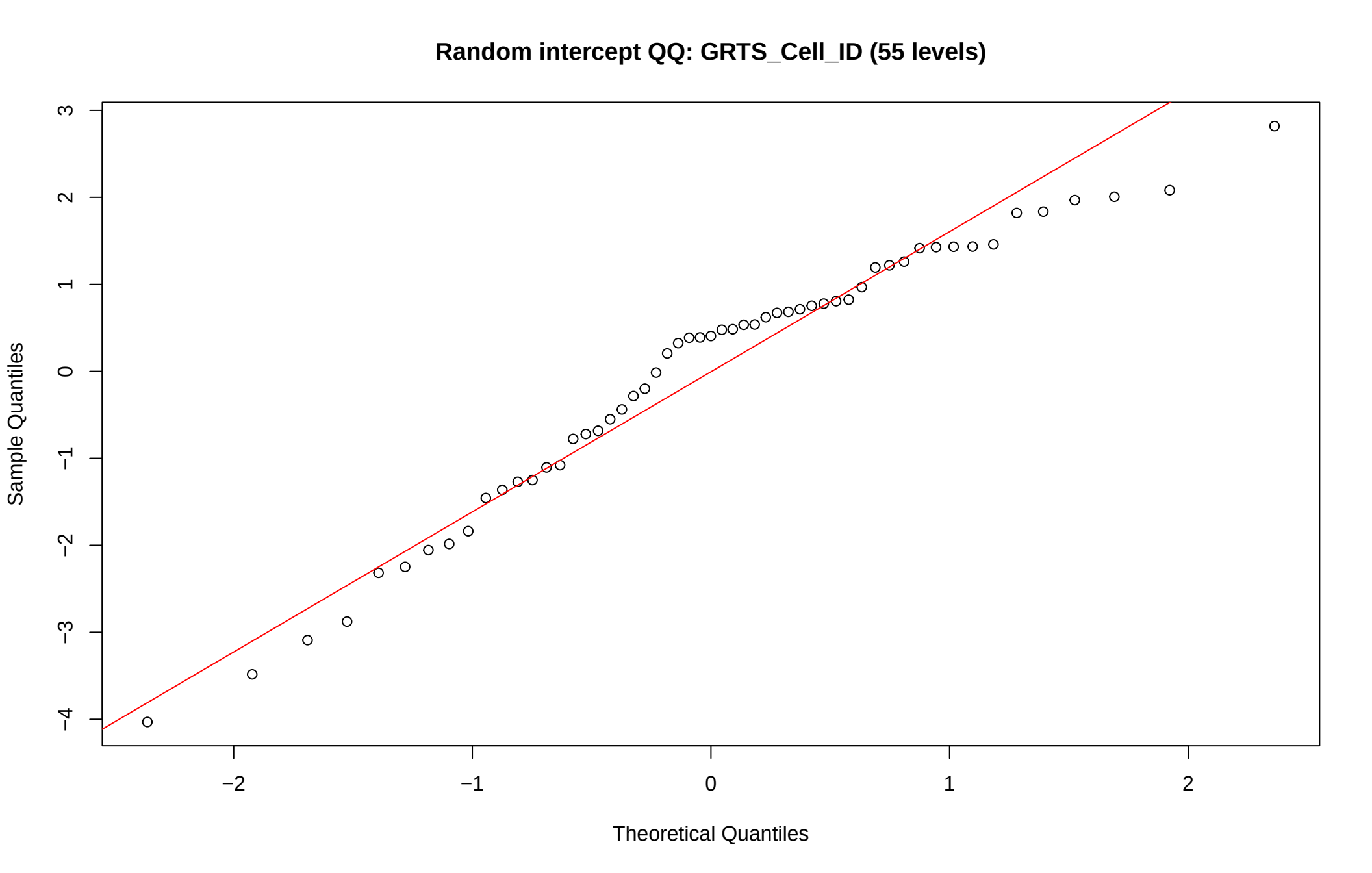

Residuals vs. time
Scaled residuals
1.0
0.8
0.6
0.4
0.2
0.0
0
2
4
6
8
Time


Autocorrelation
Durbin−Watson test p= 0.3784
Deviation n.s.
ACF
1.0
0.5
0.0
−0.5
−1.0
0
2
4
6
8
Lag


Levene Test for homogeneity of variance n.s.
Within−group deviations from uniformity significant (red)
simulationOutput$scaledResiduals
1.00
0.75
0.50
0.25
0.00
986
16426
37274
60586
79018
110378
132650
147306
194666
218282
249914
273050
grp (catPred)


Levene Test for homogeneity of variance n.s.
Within−group deviation from uniformity n.s.
simulationOutput$scaledResiduals
0.00
0.25
0.50
0.75
1.00
110378_NE
127850_SW
147306_SE2
195242_SW2
226090_NE
269777_NW
326072_NE
6202_NE
86074_SE
grp (catPred)


Random intercept QQ: Quadrant:GRTS_Cell_ID (181 levels)
Sample Quantiles
Theoretical Quantiles
3
2
1
0
−1
−2
−3
−2
−1
0
1
2


Random intercept QQ: GRTS_Cell_ID (55 levels)
Sample Quantiles
Theoretical Quantiles
3
2
1
0
−1
−2
−3
−4
−2
−1
0
1
2